SUPPORTING ADULT GUIDANCE

Calculating with bounds

Percentages, Ratio, Growth and Bounds · Lesson 9. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The lesson opens with a sofa (210 cm to the nearest 10 cm) and a gap (212 cm to the nearest cm): can we be sure it fits? The main activity, the Bounds Calculator, is in Movement. Learners choose a situation (an area, two planks end to end, the gap on a shelf, a speed), set each measurement and its degree of accuracy, and flip each value between its lower and upper bound. The exact result updates, and a result line shows all four combinations, with the current one in blue and the rounded-values result dashed. Learners can set an aim (largest or smallest) and flip until it is found, or show all four in a table. Results are exact: a division that does not terminate shows as a stacked fraction, with a labelled 2 d.p. value alongside. Six questions, true-or-not statements, a stretch on why a smaller denominator gives a larger answer, and six briefs on a second calculator follow.

Driving Question: *Which bounds give the biggest answer?*

The mathematics, in plain terms

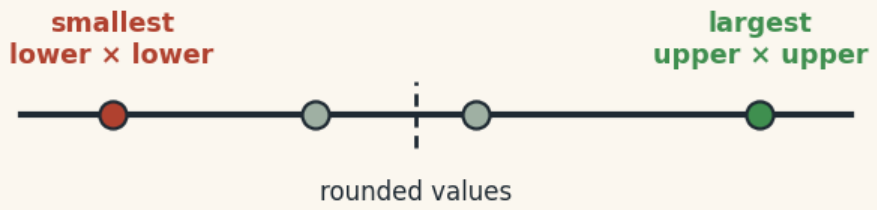
For positive values with bounds $[a_L, a_U)$ and $[b_L, b_U)$: $a + b$ and ab are increasing in both, so the extremes are LL and UU; $a - b$ and a/b are increasing in a and decreasing in b, so the largest is UL and the smallest LU. The upper bound of a result is not attained (the inputs' upper bounds are not), but by convention it is stated as the value from the bounds.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The bounds of an area

A rectangle is 5 cm by 3 cm, both to the nearest cm. Find the upper and lower bounds of its area.

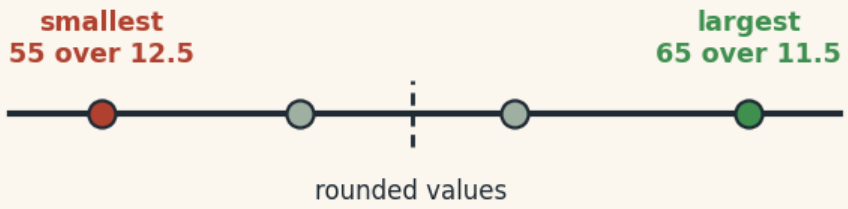


1. The bounds of each side	$4.5 \leq$ length < 5.5; $2.5 \leq$ width < 3.5
2. Largest area: upper × upper	$5.5 \times 3.5 = 19.25$
3. Smallest area: lower × lower	$4.5 \times 2.5 = 11.25$

Upper bound 19.25 cm^2; lower bound 11.25 cm^2. (The rounded values give 15 cm^2, between them.)

Example 2 — The bounds of a speed

A runner covers 60 m (to the nearest 10 m) in 12 s (to the nearest second). Find the upper and lower bounds of the speed.



1. The bounds	$55 \leq \text{distance} < 65$; $11.5 \leq \text{time} < 12.5$
2. Largest speed: upper over lower	$\frac{65}{11.5} = \frac{130}{23} = 5.65$ to 2 d.p.
3. Smallest speed: lower over upper	$\frac{55}{12.5} = 4.4$

Upper bound 5.65 m/s to 2 d.p. (exactly 130 over 23); lower bound 4.4 m/s.

Flip first, then the rules

Learners see the four possible results before any rule is stated, and look for the largest themselves. The division surprise (the smaller denominator gives the larger answer) is met on the calculator first, then explained with rope in the stretch.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Will the sofa fit? The idea card.	~5 min
Movement	The Bounds Calculator (main activity) and six questions.	~20 min
Reflection	True or not; the rope stretch.	~8 min
Creativity	Set it up: six briefs on a second calculator.	~7 min
Rest	Planning for the worst case is a kind of care.	~2 min
Nutrition	Practical: plans that work every time.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using upper – upper for the largest difference.
Ask: “To leave the most, do you take away a lot or a little?”

Using upper over upper for the largest division.
Ask: “Does dividing by a bigger number give a bigger or smaller answer?”

Calculating with the rounded values only.
Ask: “Could the length really have been a bit more?”

Using a whole unit instead of half a unit for the bounds.
Ask: “What are the bounds of this value?”

Rounding the bounds before calculating.
Ask: “Can you keep the exact bounds until the end?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: bounds of a rounded value (Lesson 8) and multiplying and dividing decimals. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says the largest gap left on an 80 cm shelf beside a 35 cm box (both to the nearest cm) is 80.5 – 35.5 = 45 cm. The aim is a question about taking away.

- “Which shelf leaves the most room?”

- “Which box takes up the least room?”
- “So which bound of the box should you take away?”
- “What is the largest gap now?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why do some answers show as fractions?
So every value is exact. A labelled 2 d.p. value is shown alongside, and rounded answers to 2 d.p. are accepted.

Why is the upper bound of the result stated, if it cannot be reached?
It is the edge of the possible results, as for a single rounded value; this is the convention in examinations.

Is the stretch required?
No. The secure core is choosing the right bound of each value for the largest and smallest answers.

What comes next?
Lesson 10: bounds in problems, deciding how sure an answer can be and to what accuracy to give it.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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